Fill in the table, with attention

Same rule as before. This time most of the table is switched off.

The reference is **TATTATGCA**. The read is **TATTTGCA**. One letter is missing from the read.

Before filling anything in, we slid the read along the reference and counted matching letters at every position. The best position told us where to look. Only the cells near it stay white. **The black cells are switched off: do not fill them in, and whenever the rule asks for a black cell, use 0.**

The rule for one cell

from the **upper-left** cell:	same letters: add 2	different letters: subtract 1
from the cell **above**:	subtract 2	
from the cell to the **left**:	subtract 2	

Write the **biggest** of the three. If the biggest is below 0, write **0**. The border, in italics, is all zeros. One cell is done for you:

- T against T: upper-left is 0, same letters, so $0 + 2 = 2$. Above gives -2. The cell to the left is black, which counts as 0, so left gives -2. Write 2.

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	2								
A	*0*									
T	*0*									
T	*0*									
T	*0*									
G	*0*									
C	*0*									
A	*0*									

When the white cells are full

1. What is the biggest number, and where is it?
2. Start there and step back to the neighbour each number came from, until you reach a 0 or the border. Draw the path.
3. One step goes **left**. That is the missing letter. Which one is it?
4. How many cells did you fill in? How many cells does the whole table have?
5. Would the whole table have given a bigger number? How could you be sure without filling it in?

Answer

With attention (only the white cells):

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	[2]	0	2						
A	*0*		[4]	2	1					
T	*0*			[6]	4	2				
T	*0*				[8]	[6]	4			
T	*0*					7	[8]	6		
G	*0*						6	[10]	8	
C	*0*							8	[12]	10
A	*0*								10	[14]

The whole table, for comparison:

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	[2]	0	2	2	0	2	0	0	0
A	*0*	0	[4]	2	1	4	2	1	0	2
T	*0*	2	2	[6]	4	2	6	4	2	0
T	*0*	2	1	4	[8]	[6]	4	5	3	1
T	*0*	2	1	3	6	7	[8]	6	4	2
G	*0*	0	1	1	4	5	6	[10]	8	6
C	*0*	0	0	0	2	3	4	8	[12]	10
A	*0*	0	2	0	0	4	2	6	10	[14]

1. The biggest number is **14**, in the bottom-right cell.
2. The path is the boxed cells.
3. The step to the left skips the letter **A** of the reference (letter number 5):

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reference:  T A T T A T G C A
read:       T A T T _ T G C A
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4. 23 cells, out of 72 in the whole table.
5. No. The whole table also gives 14, and its best path is the same. Being sure without filling it in is the question for the math club: the path can drift sideways only as far as it has gaps, so a window that is wide enough and in the right place cannot miss it.